TICKER: AAPL | $268.47 ▲ +0.63% | QQQ | $609.74 ▲ +0.54%
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APPLE INC. - SIMPLE LINEAR REGRESSION

FIN 617 – TOOLS OF FIN ANALYTICS

DePaul University – Kellstadt Graduate School of Business

By:

Punit Alagappan Kasi

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Regression Analysis: Apple (AAPL) vs. Nasdaq-100 (QQQ) Weekly Returns

Introduction and Objective

The purpose of this project is to analyze the relationship between Apple Inc. (AAPL) stock returns, and the Nasdaq-100 index which is the QQQ ETF. As a primary component of the Nasdaq-100, Apple's performance is expected to closely track the market. My hypothesis is that Apple's weekly log returns are positively and significantly explained by the Nasdaq-100's weekly returns. This analysis aims to quantify that relationship and determine how much of Apple's volatility can be explained by the market movement using regression analysis.

Data Collection and Preparation

The historical data for Apple (AAPL) and QQQ were sourced using the Excel STOCKHISTORY function, which retrieves adjusted closing price from Yahoo Finance.

Adjusted close prices reflect total return, including dividends and splits. Data were collected from 2019 through November 2025. Daily data were converted into weekly data by selecting the final trading day of each week, typically Friday, to maintain consistency across both series.

Weekly log returns were calculated using the formula:

$r_t = \ln(P_t / P_{t-1})$

Where P_t represents the adjusted closing price at the end of week t. The datasets were merged by date, ensuring each week contained corresponding return for both AAPL and QQQ. The final dataset consisted of 157 paired weekly returns.

Regression Model and Equation

The regression equation used in this analysis is based on a simple linear regression model:

$r_AAPL,t = \beta_0 + \beta_1 \times r_QQQ,t + \varepsilon_t$

Where:

- r_AAPL,t = Apple's weekly log return (dependent variable)
- r_QQQ,t = Nasdaq-100 weekly log return (independent variable)
- β_0 = intercept (alpha), representing Apple's average excess return not explained by the market
- β_1 = slope (beta), representing Apple's sensitivity to market movements
- ε_t = random error term

using Excel's data Analysis Toolpak, the following regression equation was estimated:

r_AAPL = -0.00377 + 0.93487 × r_QQQ

This means that a 1% increase in the Nasdaq-100's return corresponds to an average 0.93487% increase in Apple's return. The intercept (-0.00377) implies a small negative weekly alpha, which is economically negligible.

Regression Results and Interprertation

The regression results show that Apple's returns are strongly correlated with market returns. The R^2 value of 0.467 indicates that approximately 46.7% of Apple's weekly return variation is explained by Nasdaq-100 movements. This is a meaningful level of explanatory power, especially for a single-factor model that relies only on one independent variable (market return), which means nearly half of Apple's ups and downs in the market which is directly corelated to what happening in the broader market, while remaining 53.3% variation comes from Apple specific influences such as new product launches, earnings announcements, supply-chain updates, or changes in investor sentiment, which is also called idiosyncratic risk and which is controllable and manageable in nature, we can even denote that the price movements can be structured by Apple.

The slope coefficient ($\beta_1 = 0.935$) measures Apple's sensitivity to market movements. Because β_1 is slightly above 1, it implies that Apple behaves in a slightly on the less risker side of stocks by .065, which tends to move less than the market on a good day for the market, if QQQ rises by 1%, Apple's stock is expected on average to rise by approximately 0.935% and it is also likely to decline by around 0.935% if Nasdaq falls by 1%. This aligns with Apple stock being a large cap technology company which often amplifies broader market momentum, and a slightly defensive stock than the market.

Intercept ($\beta_0 = -0.003775$) represents Apple's average weekly return when the market's return is zero. The negative sign suggests that, when the Nasdaq-100 is flat, Apple's expected return is around -0.38% per week. However, the magnitude of the coefficient is economically very small, and its statistical insignificance means it is not quite far from zero. There is no historical evidence of systematic excess return once market performance is taken into consideration, To the conclusion that Apple is a perfect example of a larger cap stock which tracks the markets rather than outperforms it.

The t-statistic of 11.65 for β_1 is extremely large in absolute value, which indicates overwhelming evidence that the relationship between Apple and the Nasdaq-100 is real and not random. Testing the hypothesis to that, we reject the null hypothesis H_0: $\beta_1 = 0$, meaning Apple's slope coefficient is statistically significant at well beyond the 1% confident level. The corresponding p-value of is effectively 0, confirming that the connection between hold and is true.

To verify this result at the overall model level, the ANOVA F-statistic (192.44) assesses the joint explanatory power of the regression. The F-statistics compare how much of the total variation in Apple's returns is explained by the model (SSR) VS how much remains unexplained (SSE). A higher F-value indicates a stronger, statistically significant model. In this case, the F-value is exceptionally large and corresponds exactly to the squared t-statistic ($11.65^2 \approx 135.73$), which serves as a useful cross-check that both measures of significance are consistent.

And for the end where I compare $F = t^2$ where we can confirm that the data is accurate for a one-variable regression. The negligible difference here ($F - t^2 \approx -2.842E$) only states there is a rounding error as it's a big set of data's and not from any substantive discrepancy which is all in the appendix v.

To conclude, the model is statistically sound and economically meaningful. Apple stock clearly exhibits a clear, measurable and highly significant relationship with the Nasdaq-100 index. The slope coefficient shows Apple closely relates to the market movements, while the high R^2 and great test statistics validate that this linear relationship captures a substantial portion of Apple's return behavior. Lastly the remaining unexplained variation highlights the inherent company-specific risk that can't be diversified away from using market exposure alone as shown in the diagram.

Mean Reversion Analysis

This test is conducted on Apple's weekly returns to determine whether the stock exhibits a tendency to revert towards a long run equilibrium level. Mean revision assumes that an asset's returns fluctuate around a stable mean, rising when above it and falling when above it

The formula being:

$r_t = b_0 + b_1 r_{t-1} + \varepsilon_t$

Here, b0 represents the average weekly drift, b1 measures the degree of return persistence, and the resulting long run mean $\bar{r} = \frac{b_0}{1 - b_1}$ captures the equilibrium level Apple's returns naturally gravitate toward. The half-life of reversion measures how quickly deviations decay, indicating the speed with which the series returns to equilibrium after a shock.

Using the regression output, Apple's mean reverting level is around 0.00386, meaning the stock tends to stabilize around 0.386% per week, or roughly 20-21% annualized when compounded. This value reflects the long-term return tendency embedded in Apple's price dynamics over the sample period.

The slope coefficient b1 = -0.1169 is crucial here, the negative value confirms strong mean reversion: returns tend to reverse direction rather than persist. Because $|b_1| < 1$, the process satisfies the condition for covariance stationarity which also ensures that return series is stable and does not drift uncontrollably over time. A negative slope of this magnitude means shocks do not accumulate, they do recover and unwind quickly

The interpretation is reinforced by half-life of 0.32 weeks, which indicates what any deviation in Apple's stock is cut in half in roughly 2 to 3 days. For a large-cap and a high liquid stock like Apple, the speed of recovery and consistency is on par with the market expectations, information is absorbed quickly which states that the market does take into information more than emotions for this stock and price adjusts efficiently.

Overall, the mean revision shows analysts that mega stocks like Apple which have stable returns dynamic. Short-term over-reactions in weekly returns correct themselves almost immediately, leaving little room for mispricing and opportunities. However, this predictable reversion pattern can be exploited by quantitative and market-neutral strategies that capitalize on swift corrections toward the long-run mean, particularly in the highly active tech sector, refer to the appendix VIII for detailed data understanding.

Conclusion

Well Apple in this analysis demonstrates to us that it moves very closely with the Nasdaq-100 rise or fall, Apple tends to follow in the same direction and with almost the intensity, by this we understand that Apple as in Apple demonstrates true like a leader in tech sector and tied to broader market environment. However not everything is explained by the market about Apple. The remaining movements come from company specifically as show by out slope which tells us most of, I mean more than 50% comes from company specific events such as product announcements, earnings reports or investors reactions.

The overall regression model works well.

- $R^2 = 0.467$ → The Nasdaq-100 explains almost half of Apple's weekly movements.
- SEE = 0.0266 → The average prediction error is small.
- F-stat = 135.878 → The model is highly significant.

To conclude, this study confirms the original expectation: Apple returns are strongly connected to the broader tech market, and the part that is not market-driven tends to stabilize quickly. Apple's behavior is exactly how a major, well established technology company must and should, highly responsive to market forces, quick to absorb new information and stable over the long run. This confirms the original hypothesis and shows that Apple's return behavior is both statistically and economically meaningful.

Appendices

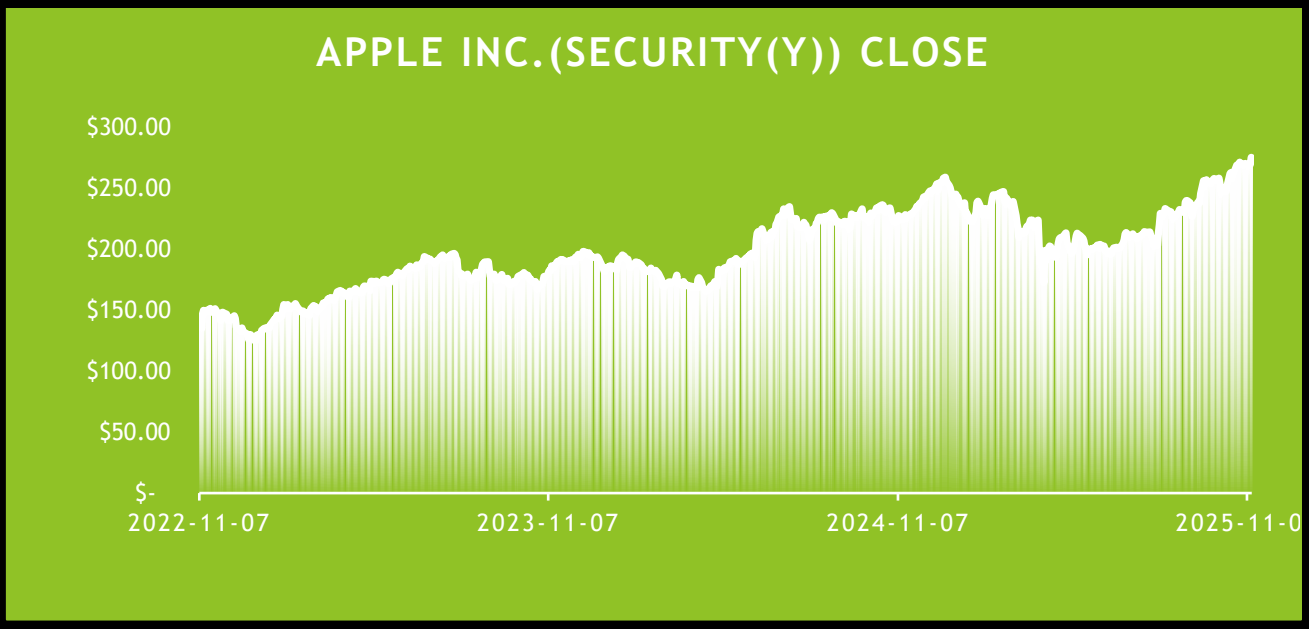
Appendix I. Weekly returns of Apple for 3 years (Security Y)

AAPL_WeekEnd	AAPL_WeekClose	AAPL_weekly_r
2024-11-08 00:00:00	226.96	0
2024-11-15 00:00:00	225	-0.00867
2024-11-22 00:00:00	229.87	0.021414
2024-11-29 00:00:00	237.33	0.031938
2024-12-06 00:00:00	242.84	0.022951
2024-12-13 00:00:00	248.13	0.02155
2024-12-20 00:00:00	254.49	0.025309
2024-12-27 00:00:00	255.59	0.004313
2025-01-03 00:00:00	243.36	-0.04903
2025-01-10 00:00:00	236.85	-0.02711
2025-01-17 00:00:00	229.98	-0.02943
2025-01-24 00:00:00	222.78	-0.03181
2025-01-31 00:00:00	236	0.057647
2025-02-07 00:00:00	227.63	-0.03611
2025-02-14 00:00:00	244.6	0.071903
2025-02-21 00:00:00	245.55	0.003876
2025-02-28 00:00:00	241.84	-0.01522
2025-03-07 00:00:00	239.07	-0.01152
2025-03-14 00:00:00	213.49	-0.11317
2025-03-21 00:00:00	218.27	0.022143
2025-03-28 00:00:00	217.9	-0.0017
2025-04-04 00:00:00	188.38	-0.14558
2025-04-11 00:00:00	198.15	0.050563
2025-04-18 00:00:00	196.98	-0.00592

2025-04-25 00:00:00	209.28	0.060571
2025-05-02 00:00:00	205.35	-0.01896
2025-05-09 00:00:00	198.53	-0.03378
2025-05-16 00:00:00	211.26	0.062149
2025-05-23 00:00:00	195.27	-0.07871
2025-05-30 00:00:00	200.85	0.028175
2025-06-06 00:00:00	203.92	0.015169
2025-06-13 00:00:00	196.45	-0.03732
2025-06-20 00:00:00	201	0.022897
2025-06-27 00:00:00	201.08	0.000398
2025-07-04 00:00:00	213.55	0.060168
2025-07-11 00:00:00	211.16	-0.01125
2025-07-18 00:00:00	211.18	9.47E-05
2025-07-25 00:00:00	213.88	0.012704
2025-08-01 00:00:00	202.38	-0.05527
2025-08-08 00:00:00	229.35	0.125102
2025-08-15 00:00:00	231.59	0.009719
2025-08-22 00:00:00	227.76	-0.01668
2025-08-29 00:00:00	232.14	0.019048
2025-09-05 00:00:00	239.69	0.032006
2025-09-12 00:00:00	234.07	-0.02373
2025-09-19 00:00:00	245.5	0.047677
2025-09-26 00:00:00	255.46	0.039769
2025-10-03 00:00:00	258.02	0.009971
2025-10-10 00:00:00	245.27	-0.05068
2025-10-17 00:00:00	252.29	0.02822
2025-10-24 00:00:00	262.82	0.04089
2025-10-31 00:00:00	270.37	0.028322

2025-11-07 00:00:00	268.47	-0.00705
2025-11-14 00:00:00	275.25	0.025

Appendix II. Apple's Historic pricing Chart



Appendix III. Weekly Returns of NASDAQ-100 ETF for 3years (Predictor(x)) and Raw data

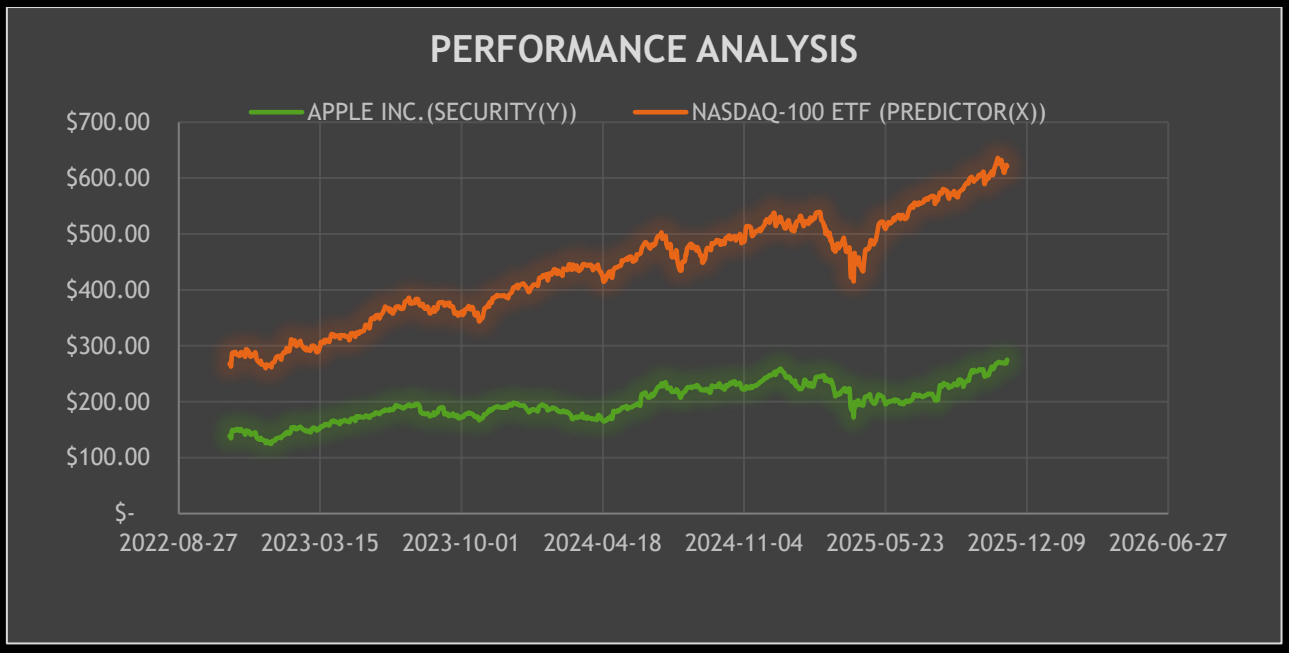
QQQ_WeekEnd	QQQ_WeekClose	QQQ_weekly_r
2022-11-11	287.96	-0.011
2022-11-18	284.82	0.007
2022-11-25	286.92	0.019
2022-12-02	292.55	-0.037
2022-12-09	282.04	-0.028
2022-12-16	274.25	-0.025
2022-12-23	267.36	-0.004
2022-12-30	266.28	0.009
2023-01-06	268.8	0.044
2023-01-13	280.97	0.006
2023-01-20	282.68	0.047
2023-01-27	296.26	0.033
2023-02-03	306.18	-0.021
2023-02-10	299.7	0.005
2023-02-17	301.16	-0.031
2023-02-24	291.85	0.026
2023-03-03	299.68	-0.038
2023-03-10	288.55	0.057
2023-03-17	305.36	0.018
2023-03-24	310.89	0.032
2023-03-31	320.93	-0.009
2023-04-07	318.05	0.002
2023-04-14	318.57	-0.006
2023-04-21	316.61	0.019

2023-04-28	322.56	0.001
2023-05-05	322.89	0.007
2023-05-12	325.03	0.035
2023-05-19	336.51	0.035
2023-05-26	348.4	0.018
2023-06-02	354.65	0
2023-06-09	354.5	0.037
2023-06-16	367.93	-0.015
2023-06-23	362.54	0.019
2023-06-30	369.42	-0.009
2023-07-07	366.24	0.034
2023-07-14	379.07	-0.009
2023-07-21	375.63	0.021
2023-07-28	383.48	-0.03
2023-08-04	372.04	-0.016
2023-08-11	366.24	-0.022
2023-08-18	358.13	0.016
2023-08-25	364.02	0.037
2023-09-01	377.59	-0.013
2023-09-08	372.58	-0.005
2023-09-15	370.81	-0.035
2023-09-22	357.91	0.001
2023-09-29	358.27	0.018
2023-10-06	364.7	0.002
2023-10-13	365.28	-0.03
2023-10-20	354.6	-0.027
2023-10-27	345.31	0.063
2023-11-03	367.71	0.029
2023-11-10	378.39	0.02
2023-11-17	386.04	0.009
2023-11-24	389.51	0.001
2023-12-01	389.94	0.006
2023-12-08	392.17	0.033
2023-12-15	405.34	0.007
2023-12-22	408.38	0.003
2023-12-29	409.52	-0.032
2024-01-05	396.75	0.032
2024-01-12	409.56	0.028
2024-01-19	421.18	0.006
2024-01-26	423.81	0.012
2024-02-02	429.01	0.019
2024-02-09	437.05	-0.015
2024-02-16	430.57	0.014
2024-02-23	436.78	0.02
2024-03-01	445.61	-0.015
2024-03-08	439.02	-0.012
2024-03-15	433.92	0.028
2024-03-22	446.38	-0.005
2024-03-29	444.01	-0.008
2024-04-05	440.47	-0.005
2024-04-12	438.27	-0.055
2024-04-19	414.65	0.039
2024-04-26	431	0.01

2024-05-03	435.48	0.015
2024-05-10	442.06	0.022
2024-05-17	451.76	0.014
2024-05-24	457.95	-0.016
2024-05-31	450.71	0.027
2024-06-07	462.96	0.034
2024-06-14	479.19	0.002
2024-06-21	480.18	-0.002
2024-06-28	479.11	0.035
2024-07-05	496.16	-0.003
2024-07-12	494.82	-0.04
2024-07-19	475.24	-0.026
2024-07-26	462.97	-0.031
2024-08-02	448.75	0.004
2024-08-09	450.41	0.053
2024-08-16	475.03	0.01
2024-08-23	480	-0.008
2024-08-30	476.27	-0.06
2024-09-06	448.69	0.058
2024-09-13	475.34	0.015
2024-09-20	482.44	0.009
2024-09-27	486.75	0.001
2024-10-04	487.32	0.012
2024-10-11	493.36	0.002
2024-10-18	494.47	0.002
2024-10-25	495.32	-0.016
2024-11-01	487.43	0.053
2024-11-08	514.14	-0.035
2024-11-15	496.57	0.018
2024-11-22	505.79	0.008
2024-11-29	509.74	0.032
2024-12-06	526.48	0.008
2024-12-13	530.53	-0.023
2024-12-20	518.66	0.007
2024-12-27	522.56	-0.008
2025-01-03	518.58	-0.022
2025-01-10	507.19	0.028
2025-01-17	521.74	0.015
2025-01-24	529.63	-0.014
2025-01-31	522.29	0.001
2025-02-07	522.92	0.029
2025-02-14	538.15	-0.023
2025-02-21	526.08	-0.035
2025-02-28	508.17	-0.033
2025-03-07	491.79	-0.025
2025-03-14	479.66	0.002
2025-03-21	480.84	-0.025
2025-03-28	468.94	-0.104
2025-04-04	422.67	0.072
2025-04-11	454.4	-0.023
2025-04-18	444.1	0.062
2025-04-25	472.56	0.034
2025-05-02	488.83	-0.002

2025-05-09	487.97	0.066
2025-05-16	521.51	-0.024
2025-05-23	509.24	0.019
2025-05-30	519.11	0.021
2025-06-06	529.92	-0.006
2025-06-13	526.96	0
2025-06-20	526.83	0.04
2025-06-27	548.09	0.015
2025-07-04	556.22	-0.004
2025-07-11	554.2	0.013
2025-07-18	561.26	0.009
2025-07-25	566.37	-0.022
2025-08-01	553.88	0.037
2025-08-08	574.55	0.005
2025-08-15	577.34	-0.009
2025-08-22	571.97	-0.003
2025-08-29	570.4	0.01
2025-09-05	576.06	0.018
2025-09-12	586.66	0.021
2025-09-19	599.35	-0.006
2025-09-26	595.97	0.012
2025-10-03	603.18	-0.023
2025-10-10	589.5	0.024
2025-10-17	603.93	0.022
2025-10-24	617.1	0.019
2025-10-31	629.07	-0.031
2025-11-07	609.74	0.019
2025-11-14	621.57	#NUM

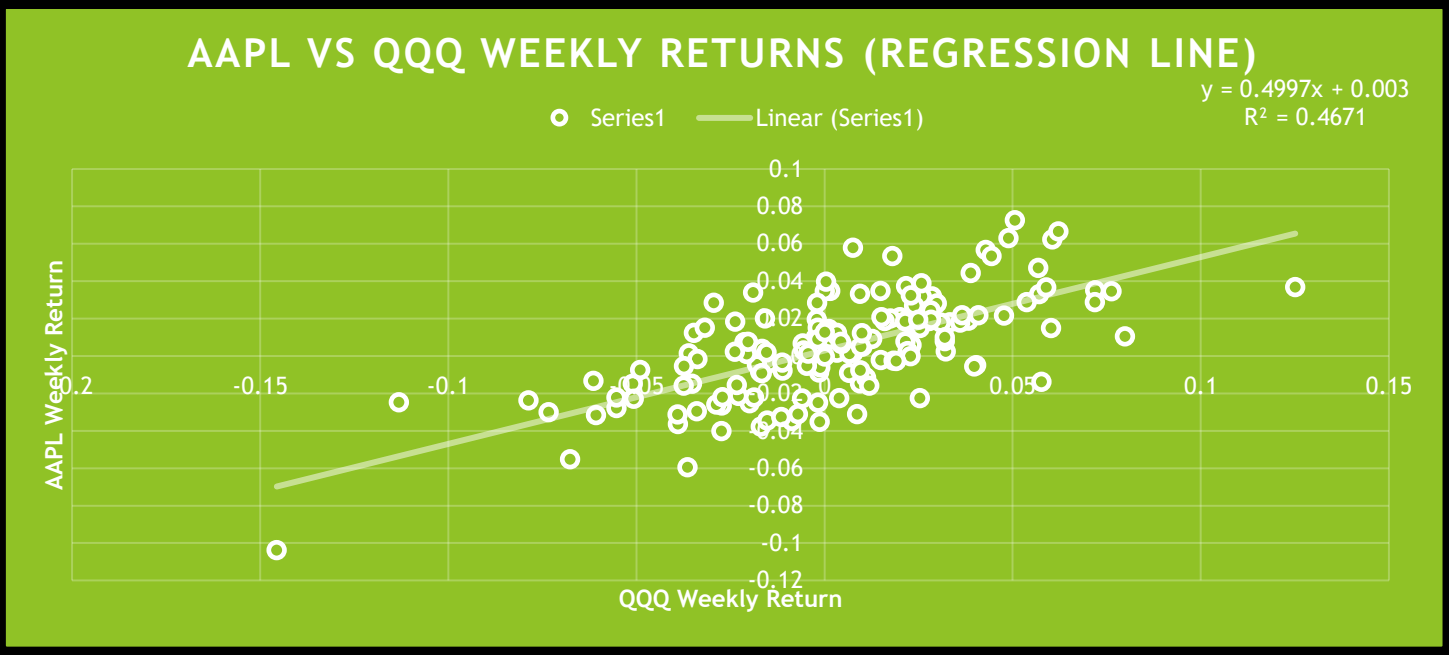
Appendix IV Performance Analysis NASDAQ 100 ETF vs APPLE INC



Appendix V. regression Analysis, T-test and F-Test Calculations

Metric	Value
N	157
Mean x (QQQ r)	0.00490081
Mean y (AAPL r)	0.00080668
Var(x)	0.0007045
Sxx = DEVSQ(X)	0.10990242
SSyy = DEVSQ(y)	0.2056244
Slope b1	0.9348731
Intercept b0	-0.003775
R^2	0.46713021
SSR	0.09605337
SSE	0.10957103
MSE = SSE/(n-2)	0.00070691
SE(b1) = SQRT(MSE/Sxx)	0.08020074
t-stat (b1)	11.6566639
p-value (two-tailed)	6.0449E-23
F-stat	135.877813

Appendix VI. Regression Line Scatter Plot



Appendix VII. ANOVA Table

ANOVA	Value
df Regression	1
df Residual	155
SSR	0.09605337
SSE	0.10957103
MSR	0.09605337
MSE	0.00070691
F = MSR/MSE	135.877813
Check: F - t^2 (should be 0)	-2.842E-13

Appendix VIII. Mean Revision Analysis

Mean Regression Analysis

Intercept (b_0)	0.00431414
Slope (b_1)	-0.1168843
Mean-Reverting Level ($\bar{x}$)	0.00386265
Half-Life of Reversion	0.32290903
Stationary?	Yes – mean-reverting